

Variation Method & FEA Course

Homework 2

Submitted before May 25, 2026

1. Identify the shortest curve between the point $A(0,0)$ and the point $B(2,1)$ by the variation method.

Hints: Let the curve be expressed as $y = y(x)$. The length functional can be written as

$$S[y] = \int_0^2 \sqrt{1 + (y')^2} dx,$$

with boundary conditions

$$y(0) = 0, \quad y(2) = 1.$$

2. Deduce the Euler equation for the extremum condition of the functional containing third-order derivative of its independent function.

3. Solve the maximum and minimum values of z on the intersection curve of the ellipsoid

$$9x^2 + 4y^2 + z^2 = 36$$

and the plane

$$2x - y + z = 3.$$

Hints: Consider the unconditional extremum of the function

$$F = z + \lambda_1(2x - y + z - 3) + \lambda_2(9x^2 + 4y^2 + z^2 - 36),$$

where λ_1 and λ_2 are Lagrange multipliers.

4. Find the function $y(x)$ leading to the extremum of the functional

$$Q[y] = \int_0^1 [(y')^2 + 4y^2 - 8xy] dx,$$

with boundary conditions

$$y(0) = 1, \quad y(1) = 2.$$

Hints: First write the Euler equation corresponding to the functional, then solve the resulting ordinary differential equation using the boundary conditions.